

Google X's balloon Internet takes a step closer to reality

Project Loon, which was launched in 2013 as part of Google X (now just X), has made a major breakthrough. According to Astro Teller, X's captain of moonshots, navigational systems that control the balloons have drastically improved beyond the wildest expectations of the project's engineers—something Teller admits happened by accident.

At first, Loon launched its balloons throwing caution to the wind, and controlling the devices just enough to catch the right air currents to take these on their way. After the team's navigation algorithms improved, flightpaths could be charted more accurately to plan return loops around oceans or continents after the balloons drifted away from their original targets, but it was still a tall task to keep these in one place.

But early last year, the Loon team noticed that some balloons lingered in one area instead of drifting away on the winds. The team identified three areas of development that helped the balloons' navigation system improve so rapidly.

One, the altitude control system was much more efficient after updating and streamlining its solar panels and air pumps, which allowed for more manoeuvres during each flight.

Second, since the balloons' algorithms depend on a lot of data to develop, reaching a point where simulated flights were just as good as real ones to refine those algorithms was a big step forward, too.

Finally, using a machine learning system to pilot the balloons was key for traversing the unpredictable conditions of the stratosphere.

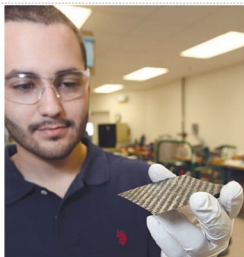


Loon's first pilot test in New Zealand in 2013; Google X (now just X), has been sending smart balloons up into the stratosphere for four years now with the aim of creating a floating, balloon-powered LTE network (Image courtesy: <https://blog.x.company>)

Batteries that could provide power to microsatellites, cubesats

Dr Luke Roberson, senior principal investigator for Flight Research within Exploration Research and Technology Directorate at NASA's Kennedy Space Center in Florida, USA, is collaborating on research of a new solid-state battery prototype with Dr Ryan Karkkainen, a composite material expert at University of Miami, USA. The battery composition was developed by Xiangyang Zhou, PhD, associate professor of mechanical and aerospace engineering. Three students from the university are currently working on the prototype with Dr Roberson.

The battery is made by heat-treating vacuum-compressed several layers of small carbon-fibre squares and placing the solid-state battery layer between these. Composite reinforcement and mechanical/electrical testing will be performed at Kennedy Space Centre in the near future. The battery is only 2mm to 3mm thick, and is suitable for use in microsatellites, including CubeSats.



The size of the battery is so small that it could be a prime candidate for use in microsatellites, including CubeSats (Image courtesy: www.nasa.gov)